

WUGANG, China

WMO: 578530

Lat: 26.7383N

Lon: 110.6370E

Elev: 340

StdP: 97.31

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.3	0.0	-7.5	2.1	4.9	-5.1	2.5	4.1	4.8	7.2	4.2	7.1	1.3	0	0.321

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.3	35.0	24.9	33.9	24.7	32.9	24.5	26.2	31.5	25.8	30.9	25.4	30.3	2.5	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.0	20.9	28.2	24.6	20.4	27.9	24.2	19.9	27.6	84.1	31.5	82.2	31.3	80.6	30.5	28.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
4.6	3.9	3.4	-2.5	36.7	1.4	1.0	-3.5	37.4	-4.3	37.9	-5.1	38.5	-6.2	39.2		
			DB	WB	DB	WB	DB	WB	DB	WB	DB	WB	DB	WB		
			-3.2	27.1	1.4	0.7	-4.2	27.5	-5.1	27.9	-5.9	28.3	-6.9	28.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.8	5.9	8.2	12.3	18.3	22.4	25.5	28.2	27.6	24.1	19.0	13.6	8.1
	DBStd	8.43	3.79	4.65	4.50	4.36	3.13	2.61	2.22	2.44	3.43	3.45	3.79	3.19
	HDD10.0	333	137	84	27	2	0	0	0	0	0	0	11	74
	HDD18.3	1422	386	286	193	54	4	0	0	0	2	33	147	316
	CDD10.0	3189	9	33	98	249	386	467	563	546	423	279	119	17
	CDD18.3	1236	0	1	7	52	131	217	305	288	176	54	6	0
	CDH23.3	11335	1	11	44	344	912	1839	3466	3000	1412	274	33	0
CDH26.7	4362	0	2	10	93	295	617	1530	1251	501	58	4	0	
Wind	WSAvg	1.4	1.3	1.4	1.4	1.5	1.5	1.4	1.6	1.6	1.4	1.2	1.2	1.2
Precipitation	PrecAvg	1380	61	73	105	173	205	195	113	136	88	104	83	45
	PrecMax	1944	131	173	204	358	384	394	236	304	276	299	275	92
	PrecMin	1067	5	11	35	55	63	60	33	11	2	5	2	1
	PrecStd	191	30	38	38	68	86	78	46	65	60	67	59	27
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.1	24.9	27.5	31.7	33.4	34.2	36.3	36.3	34.8	30.8	27.1	20.0
		MCWB	13.0	16.4	18.9	20.5	22.7	24.9	25.1	24.8	24.2	21.1	19.4	12.9
	2%	DB	16.0	20.5	24.0	29.3	31.7	32.8	35.1	35.0	33.1	28.2	23.7	17.5
		MCWB	10.9	14.4	17.6	20.8	22.0	24.8	25.1	24.7	24.0	20.3	16.4	11.5
	5%	DB	13.6	17.4	21.4	27.2	29.9	31.6	34.1	33.7	31.4	26.2	21.5	15.2
		MCWB	9.5	12.9	15.8	20.1	21.9	24.5	25.0	24.7	23.4	19.2	15.8	10.4
	10%	DB	11.5	14.7	18.9	25.0	28.2	30.2	32.9	32.4	29.6	24.5	19.6	13.1
		MCWB	8.4	11.3	14.7	19.4	21.6	24.2	24.8	24.6	22.8	18.5	14.9	9.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.6	17.1	20.5	23.5	25.5	26.5	26.9	26.7	25.7	23.7	20.5	15.3
		MCDB	17.9	23.8	26.2	28.5	29.7	31.4	32.4	32.3	31.7	28.1	25.6	17.2
	2%	WB	12.0	15.3	18.6	22.3	24.5	25.8	26.3	26.1	25.1	22.1	18.5	12.7
		MCDB	14.8	19.8	22.8	27.0	28.7	30.3	31.7	31.5	30.6	25.3	21.6	15.4
	5%	WB	10.2	13.5	17.0	21.3	23.7	25.4	25.9	25.6	24.5	20.9	17.1	11.0
		MCDB	12.7	16.5	19.8	25.6	27.4	29.5	31.2	30.8	29.3	24.2	19.9	14.2
	10%	WB	8.7	11.8	15.3	20.3	22.8	24.9	25.5	25.2	23.8	19.9	15.7	9.7
		MCDB	11.2	14.2	18.0	24.1	26.4	28.7	30.5	30.1	28.1	23.1	18.4	12.3
Mean Daily Temperature Range	5% DB	MDBR	4.9	5.5	6.0	7.1	7.0	6.2	7.3	7.3	7.2	6.6	6.7	6.0
		MCDBR	10.6	11.6	11.8	11.3	10.9	8.7	9.7	9.8	10.2	10.2	11.0	11.7
		MCWBR	5.8	6.0	5.4	4.4	3.8	2.8	2.6	2.5	3.1	3.6	4.8	5.9
	5% WB	MCDBR	8.9	10.0	9.8	9.5	7.9	7.1	8.0	8.0	8.4	7.5	8.3	9.3
		MCWBR	5.2	5.7	5.4	4.6	3.5	2.7	2.7	2.6	2.9	3.7	4.4	5.3
Clear-Sky Solar Irradiance	taub	0.665	0.691	0.766	0.734	0.637	0.615	0.557	0.639	0.674	0.731	0.652	0.646	
	taud	1.522	1.519	1.436	1.538	1.764	1.833	1.996	1.760	1.659	1.499	1.607	1.574	
	Ebn at Noon	573	607	595	637	704	715	757	693	650	570	581	563	
	Edh at Noon	251	271	310	286	228	211	179	226	244	272	229	230	
All-Sky Solar Radiation	RadAvg	1.59	1.94	2.48	3.16	3.58	3.68	4.86	4.37	3.62	2.88	2.38	1.90	
	RadStd	0.38	0.50	0.34	0.47	0.61	0.53	0.53	0.36	0.39	0.39	0.41	0.45	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.54	N/A	-1.50	+0.71	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (7 neighbors)	+0.29	N/A	N/A	+0.59	N/A	N/A	N/A	N/A	+118	+71

Nomenclature: See separate page